

Ques. What is the purpose of the range() function?

- The range() function in Python is used to generate a sequence of numbers, which is often used for iterating over a loop a specific number of times.
- It is very commonly used in loops (like for loops).
- The range() function can **take** up to **three** arguments:
 - **start (optional)**: The starting value of the sequence (default is 0).
 - **stop(required)**: The stopping value of the sequence. The sequence will go up to, but not include, this value.
 - **step (optional)**: The difference between each number in the sequence (default is 1).

```
# Using range with only one argument (stop)
for i in range(5):
    print(i) # Prints 0, 1, 2, 3, 4

# Using range with two arguments (start and stop)
for i in range(2, 8):
    print(i) # Prints 2, 3, 4, 5, 6, 7

# Using range with all three arguments (start, stop, step)
for i in range(1, 10, 2):
    print(i) # Prints 1, 3, 5, 7, 9
```